Research Article

Coach WAR: A Counterfactual Approach to Evaluating NFL Head Coach Impact in the Modern Era

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Abstract: Evaluating NFL head coach performance is challenging due to complex interactions among coaching decisions, player talent, and organizational factors. We introduce a counterfactual framework measuring head coach impact by predicting team performance under a replacement-level coach, then comparing this baseline to actual results. This difference represents coaching Wins Above Replacement (WAR). Analyzing 1,637 team-seasons (1970–2024) with 365 features spanning draft history, injuries, rosters, team performance, and coaching histories, we employ XGBoost regression to isolate coaching contributions from team context. The model achieves test $R^2 =$ ~~0.602~~0.430 on held-out coaches, with a residual standard error of ±~~2.1~~2.5 wins per season. Top-ranked coaches average up to +2.3 WAR per season (95% CI: [~~+1.3~~+1.4, +3.2]) while bottom-ranked coaches average ~~−3.3~~−3.1 WAR (95% CI: [~~−4.7~~−5.0, ~~−2.0~~−1.2]). Offensive versus defensive coaching backgrounds show a historical shift: ~~defensive coaches held a significant advantage in the 1970s (−0.87,~~ in the 1970s defensive coaches outperformed offensive coaches by 0.91 wins per season (95% CI: [~~−1.45~~−1.42, ~~−0.29~~−0.39], $p =$ ~~0.005~~0.0003), while offensive coaches averaged +~~0.59~~0.46 more in the 2020s (95% CI: [~~−0.05~~−0.17, ~~+1.24~~+1.09], $p =$ ~~0.11~~0.22). ~~Year-to-year persistence is low ($R^2 = 0.069$, $p < 0.001$): top-quintile coaches lose approximately 79% of their WAR advantage by the following season, consistent with scheme obsolescence.~~Year-to-year persistence is low ($R^2 = 0.064$): a single season mixes a small but genuine coaching signal with substantial sampling noise.

Keywords: coaching analytics, wins above replacement, NFL, sports analytics

1 Introduction

Evaluating head coach performance in the National Football League (NFL) presents ~~unique~~ challenges due to complex interactions among coaching decisions, player talent, and organizational factors. Traditional metrics such as win–loss records conflate team success with coaching quality, failing to isolate a coach's contribution from the context in which that coach operates.

We propose a counterfactual framework for measuring head coach impact: for each team-season, we predict how the team would have performed under a replacement-level coach with league-median characteristics, then compare this prediction to actual results. The difference represents coaching Wins Above Replacement (WAR). By controlling for roster quality, strength of schedule, salary allocation, injury exposure, and other contextual factors, the framework isolates the coaching contribution from the surrounding talent and organizational context.

2 Literature Review

Prior research on coaching evaluation has taken several approaches. Hadley et al. (2000) used a Poisson regression framework with basic team statistics to assess NFL coaching efficiency relative to production frontiers, while Goff and Wisley (2006) examined managerial lifecycle effects. ~~The present work expands on~~

We expand on Hadley et al. (2000) by isolating coaching impact from a broader set of contextual factors, including salary distribution, player turnover, injury data, and detailed coach-specific performance histories.

WAR methodologies for player evaluation are well established in baseball (Palmer and Thorn, 1984; Smith, 2008; Woolner, 2002) and have been extended with formal uncertainty quantification. Baumer et al. (2015) developed openWAR with resampling-based interval estimates, advocating for uncertainty quantification as standard practice in WAR metrics. Brill and Wyner (2024) proposed Grid WAR as an alternative aggregation for pitchers, further demonstrating the importance of methodological choices in WAR construction. In football, Yurko et al. (2019) introduced nflWAR for offensive player evaluation, and Sabin (2021) estimated player value using plus–minus models that isolate individual contributions within a team sport.

Tree-based machine learning methods have proven effective for NFL prediction tasks. Lock and Nettleton (2014) used random forests to estimate within-game win probability, demonstrating the effectiveness of ensemble tree methods for NFL prediction tasks. More broadly, the growth of NFL tracking data has expanded the scope of quantitative football research (Lopez, 2020). In basketball, Deshpande and Jensen (2016) estimated individual player impact on team winning probability, addressing challenges of attribution in team sports that parallel those in coaching evaluation.

Our contribution extends the WAR framework to coaching evaluation, adapting the counterfactual approach to isolate coaching impact in a setting where coaches influence outcomes indirectly through strategic and personnel decisions rather than direct play execution.

3 Methods

3.1 Data Sources and Feature Engineering

We analyzed 1,637 team-seasons spanning 1970–2024, representing all 32 current NFL franchises across the Modern Era of professional football. The analysis incorporated 365 features from multiple data sources: draft pick history, injury reports, starter designations, complete rosters, historical team performance, and comprehensive head coach career histories from Pro Football Reference, along with salary cap allocation and contract details from Spotrac. Feature counts by category are shown in Appendix A (Table 6).

The coach-specific feature set comprised 139 features in three groups: (1) seven coaching experience variables (years of NFL and college experience as position coach, coordinator, or head coach, plus number of prior head coaching hires); (2a) 33 normalized team performance statistics from the coach's most recent offensive coordinator tenure, and (2b) 33 from the most recent defensive coordinator tenure. Because no coach in our sample served as both offensive and defensive coordinator prior to becoming a head coach, exactly one of these two groups is always absent for each coach; and (3) 66 normalized statistics from the coach's head coaching history, comprising 33 team offensive metrics and 33 opposing-team offensive metrics (i.e., defensive performance). All performance features were z-score normalized within each season to ensure cross-era comparability. Non-coaching features captured roster construction (turnover rates, player age and experience by position), financial management (salary cap allocation percentages), injury exposure (games missed by starters), draft capital investment, and team performance context (strength of schedule, penalty rates). Data availability varied by category: team performance and roster metrics extended across all seasons, while salary cap data was limited to 2011–2024 and player injury data to 2009–2024.

3.2 The Counterfactual Framework

To isolate coaching impact from roster quality, organizational context, and other factors, we developed a counterfactual approach: for each team-season, we predict what the team's performance would have been under a replacement-level coach with league-median coaching characteristics, then compare this prediction to actual results.

Formally, let y_i denote the actual regular-season winning percentage for team-season i, and let $\mathbf{x}_i = (\mathbf{x}_i^{(c)}, \mathbf{x}_i^{(t)})$ denote the corresponding feature vector ~~for team-season i~~, where $\mathbf{x}_i^{(c)} \in \mathbb{R}^{139}$ are coaching features and $\mathbf{x}_i^{(t)} \in \mathbb{R}^{226}$ are non-coaching features. We compute replacement-level coaching features $\tilde{\mathbf{x}}^{(c)}$ as follows: for each coach j, we first compute the career mean $\bar{\mathbf{x}}_j^{(c)}$ across all of coach j's seasons, then take the component-wise median across all coaches:

$$\tilde{x}_k^{(c)} = \mathrm{median}_j\big(\bar{x}_{j,k}^{(c)}\big), \quad k \in \{1, \dots, 139\}. \tag{1}$$

This two-stage aggregation ensures each coach is weighted equally regardless of tenure length. The replacement-level prediction is then $\hat{y}_i^{(r)} = f(\tilde{\mathbf{x}}^{(c)}, \mathbf{x}_i^{(t)})$, where f is the trained model.

This approach parallels established WAR methodologies in baseball (Baumer et al., 2015; Palmer and Thorn, 1984; Woolner, 2002), football (Yurko et al., 2019), and basketball (Berri et al., 2006; Deshpande and Jensen, 2016), where player contributions are measured against replacement-level alternatives. Unlike baseball's replacement level, set at a .294 winning percentage representing readily available minor league talent (Appelman and Cameron, 2013; Baseball-Reference.com, n.d.), we define replacement-level coaching as the league median. ~~This choice reflects the coaching labor market's structure: NFL head coaches average only 3.2 years of tenure (Goff and Wisley, 2006), substantially shorter than MLB players' average career of 5.6 years (Witnauer et al., 2007), making the median coach a more appropriate baseline than a theoretical minimum-quality hire.~~ Two considerations motivate this choice. First, the median represents the realistic options a franchise faces when filling a head-coaching vacancy: an experienced coordinator, former head coach, or top college coach drawn from the existing pool of NFL-caliber candidates. A low-percentile baseline would model a replacement that is far less commonly hired in practice. Second, the median is robust to outliers in the coach distribution, so a few exceptional or exceptionally poor coaching careers do not distort the baseline.

3.3 Model Specification

We employed XGBoost (Chen and Guestrin, 2016), a gradient-boosted regression tree ensemble that minimizes the regularized objective

$$\mathcal{L} = \sum_{i=1}^{n} \ell\big(y_i, \hat{y}_i\big) + \sum_{m=1}^{M} \Omega(f_m), \quad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda \|w\|^2 + \alpha \|w\|_1, \tag{2}$$

where ℓ is the squared-error loss, T is the number of leaves in tree f, w are leaf weights, and γ, λ, α are regularization parameters. XGBoost was selected for its ability to capture non-linear relationships and feature interactions while providing interpretable feature importance metrics, properties also leveraged in NFL analytics by Lock and Nettleton (2014) for win probability estimation.

Missing values arising from incomplete early-era salary and injury data were imputed via truncated Singular Value Decomposition (SVD) matrix completion (Brand, 2002). The decomposition iterated until convergence (tolerance 10^{-4}), and non-normalized features were standardized prior to imputation to prevent scale-dependent bias. To prevent train–test contamination through the imputation step, validation imputation is performed *separately* on the training and test subsets (Pipeline 1, below); the standardization is fit on the training subset and applied to the test subset.

We adopt a two-pipeline architecture that separates model validation from coach WAR estimation. *Pipeline 1 (validation)* measures honest out-of-sample generalization with the split-then-impute procedure described above. *Pipeline 2 (estimation)* refits the same XGBoost specification on all 1,637 observations after a single full-data SVD imputation, maximizing data efficiency for coach-level WAR estimates. The reported single-season WAR standard error (Section 3.5.1) is taken from Pipeline 1's test RMSE so that uncertainty reflects honest generalization rather than in-sample fit. Hyperparameters are selected once via Pipeline 1's training set and held fixed for both pipelines.

To prevent data leakage, we employed a coach-based train/test ~~stratification~~split with no further stratification: 216 coaches and all their associated seasons were randomly assigned to the training set (81.7% of observations), while 54 randomly held-out ~~held-out~~ coaches formed the test set (18.3%); the assignment used a fixed random seed (42) for reproducibility. This ensures the model's test performance reflects generalization to entirely unseen coaching tenures. Hyperparameters were selected via randomized search over a discrete grid with 200 iterations and 5-fold cross-validation on the training set, optimizing R^2. Final hyperparameter values are shown in Appendix B (Table 7). To verify that the chronological structure of the dataset (1970–2024) does not bias the random coach split, we additionally tested a chronological cutoff (training on Year $\leq$ 2014, testing on Year $\geq$ 2015) combined with split-then-impute, yielding test $R^2 = 0.443$ — essentially identical to the random coach split's $R^2 = 0.430$ — indicating that within-year normalization of the coaching performance features absorbs era-mixing effects.

3.4 WAR Calculation

Coach WAR is computed as the difference between actual winning percentage and the replacement-level prediction, scaled to a 16-game season:

$$\mathrm{WAR}_i = 16 \times \left(y_i - \hat{y}_i^{(r)}\right), \tag{3}$$

where y_i is the actual winning percentage and $\hat{y}_i^{(r)} = f(\tilde{\mathbf{x}}^{(c)}, \mathbf{x}_i^{(t)})$ is the replacement-level prediction from Equation 1. All WAR calculations are standardized to a 16-game season, including the 17-game seasons beginning in 2021, to ensure cross-era comparability and prevent artificial inflation of recent values. For 17-game seasons, this represents a 6% conservative adjustment.

3.5 Uncertainty Quantification

As advocated by Baumer et al. (2015) for player WAR, we report interval estimates alongside point estimates. Because coaching WAR is observed at the season level without sub-seasonal granularity, we use analytical t-intervals rather than the bootstrap resampling approach used for play-level player metrics. All p-values are two-sided unless otherwise noted.

3.5.1 Single-Season Precision

We report the model residual standard error, computed as the root mean squared error on the held-out coach test set converted to a 16-game season scale:

$$\mathrm{SE}_{\mathrm{model}} = 16 \times \mathrm{RMSE}_{\mathrm{test}}. \tag{4}$$

This provides a global measure of single-season WAR precision applicable to all observations. Because XGBoost does not produce observation-level prediction intervals, we treat $\mathrm{SE}_{\mathrm{model}}$ as a uniform bound on single-season estimation uncertainty.

3.5.2 Career WAR Intervals

For coach c with n_c seasons of individual WAR values $\{w_{c,1}, \ldots, w_{c,n_c}\}$, the career average WAR and its 95% confidence interval are

$$\overline{w}_c \pm t_{n_c-1,\,0.025} \cdot \frac{s_c}{\sqrt{n_c}}, \tag{5}$$

where s_c is the sample standard deviation of individual-season WAR values and $t_{n_c-1,\,0.025}$ is the critical value from the t-distribution with $n_c - 1$ degrees of freedom. These intervals directly address career sample size: coaches with few seasons (e.g., $n_c = 3$) produce wide intervals, while long-tenured coaches yield narrow intervals.

3.5.3 Decade Comparisons

To assess whether offensive or defensive coaching backgrounds produce systematically different WAR outcomes, we compare group means within each decade using two complementary tests. First, Welch's t-test (unequal variances) yields a t-statistic and p-value for the null hypothesis that the two group means are equal. The 95% confidence interval on the difference in means is

$$(\bar{w}_O - \bar{w}_D) \pm t_{\nu,\,0.025} \cdot \sqrt{\frac{s_O^2}{n_O} + \frac{s_D^2}{n_D}}, \tag{6}$$

where subscripts O and D denote offensive and defensive background groups, and ν is the Welch–Satterthwaite approximation to degrees of freedom:

$$\nu = \frac{\left(\frac{s_O^2}{n_O} + \frac{s_D^2}{n_D}\right)^2}{\frac{(s_O^2/n_O)^2}{n_O-1} + \frac{(s_D^2/n_D)^2}{n_D-1}}. \tag{7}$$

Second, the Mann–Whitney U test provides non-parametric confirmation by testing whether one group's WAR values tend to rank higher than the other's, without distributional assumptions. Effect size is reported as Cohen's d using the pooled standard deviation.

To quantify whether the offensive-versus-defensive gap has shifted over time, we fit a separate ordinary least squares regression of the per-decade mean difference ($\bar{w}_O - \bar{w}_D$) on decade midpoint. The slope estimates the change in the gap per decade (in games per season), ~~and its p-value tests whether the trend is significantly different from zero.~~ and its p-value quantifies the strength of evidence against the null hypothesis of zero slope.

3.5.4 Persistence Analysis

To evaluate year-to-year stability of coaching performance, we compute Pearson and Spearman correlation coefficients between consecutive-season WAR values within each coaching background group. To test whether persistence differs between offensive and defensive coaches, we apply Fisher's Z-transformation:

$$Z = \frac{\frac{1}{2}\ln\left(\frac{1+r_O}{1-r_O}\right) - \frac{1}{2}\ln\left(\frac{1+r_D}{1-r_D}\right)}{\sqrt{\frac{1}{n_O-3} + \frac{1}{n_D-3}}}, \tag{8}$$

where r_O and r_D are the Pearson correlations for the offensive and defensive groups, n_O and n_D are the respective sample sizes, and Z is compared to a standard normal distribution.

3.5.5 Multiple Testing Adjustment

Because this analysis evaluates multiple inferential hypotheses, we apply the Benjamini–Hochberg (BH) procedure (Benjamini and Hochberg, 1995) to control the false discovery rate (FDR) across the family of reported tests. Following Wasserstein and Lazar (2016), we de-emphasize threshold-based dichotomies and instead report effect sizes, confidence intervals, and exact p-values, treating any adjusted threshold as one input to interpretation rather than a categorical rule. The family comprises 13 tests: six per-decade Mann–Whitney U tests for offensive–defensive WAR differences, two Welch's t-tests comparing 1970s with 2020s mean WAR within each background group, four Fisher's Z tests of correlation differences across persistence horizons (1–, 2–, 3–, and 4-year averages), and one overall Pearson test for year-to-year persistence. With $k = 13$ and a target FDR of $q^* = 0.05$, the BH procedure rejects any hypothesis whose ordered p-value satisfies $p_{(i)} \leq (i/k) \cdot q^*$, which yields a critical p-value of $p^* \approx 0.019$ for the present family. Coach-level confidence intervals reported in Tables 1 and 2 are marginal 95% intervals intended for descriptive ranking, not for pairwise hypothesis testing among coaches.

4 Results

Pipeline 1 (validation) achieved $R^2 = 0.785$ on the training set and $R^2 = 0.430$ on held-out coaches (54 of 270 coaches withheld with all their seasons). The train–test gap of 0.36 reflects the limit of feature-based prediction at the 16-game-season scale: the binomial sampling floor for a season of true win probability p is $\sqrt{16 \cdot p(1-p)} \approx 1.94$ wins, and the model's test RMSE of ± 2.5 wins per 16-game season approaches that floor. The binomial floor thus accounts for the majority of the model's residual variance: the modest single-season R^2 reflects the irreducible noise of a 16-game season rather than a failure to capture coaching signal. Because this noise is season-specific and independent from year to year, averaging across a coach's seasons cancels much of it: a career average is more precise than a single season by roughly the square root of the number of seasons observed. The multi-season rankings and confidence intervals that anchor our conclusions are therefore far more reliable than any single-season estimate.~~The final model achieved $R^2 = 0.789$ on the training set and $R^2 = 0.602$ on held-out coaches (54 of 270 coaches withheld with all their seasons). The train–test gap of 0.19 indicates moderate overfitting, mitigated by regularization (max depth = 2, ℓ_1/ℓ_2 penalties). The model residual standard error on the test set corresponds to ± 2.1 wins per 16-game season (Section 3.5), providing a baseline for interpreting single-season WAR precision.~~ To verify that our findings are not artifacts of the XGBoost algorithm, we compared coach WAR estimates to those from Ridge regression trained on the same coach-based split. The two sets of estimates are ~~correlated highly~~highly correlated (Spearman $\rho = 0.81$), confirming that relative coaching rankings are robust to modeling choice. On the same held-out coaches, however, a cross-validation-tuned Ridge regression generalizes far less well than XGBoost (test $R^2 = 0.15$ versus 0.43); the two methods therefore agree on relative rankings but differ sharply in out-of-sample accuracy, indicating that the gradient-boosted model captures predictive structure the linear model misses. The mean coaching WAR across all team-seasons was ~~−0.01~~−0.05 games (median = ~~0.06~~0.01), confirming the median replacement baseline approximately centers the distribution near zero. Feature importance analysis confirmed coaching-specific features accounted for 42% of total model importance, indicating that coaches exert substantial influence on outcomes when controlling for roster quality, financial resources, and competitive context. Appendix C (Table 8) shows the top 20 features by importance.

4.1 Football Context and Career Leaders

While single-season WAR ranged from +~~5.9~~5.7 to ~~−8.6~~−8.7 (~~Section 4.6~~Section 4.5), career averages provide more reliable assessments owing to larger sample sizes and correspondingly narrower confidence intervals. Among coaches with at least 3 seasons, the top performers averaged between ~~+1.2~~+1.1 and +2.3 WAR per season; the worst ranged from ~~−3.3~~−3.2 to ~~−1.9~~−1.8 WAR. Tables 1 and 2 report the best and worst coaches by average WAR, with 95% confidence intervals.

Long-tenured coaches accumulate enough seasons to yield precise estimates: ~~Don Shula's +2.0 WAR per season (95% CI: [+1.5, +2.5]) over 26 seasons and John Madden's +2.1 per season (95% CI: [+1.3, +2.8]) over 9 seasons both have intervals entirely above zero. Shorter careers naturally produce wider intervals. Kevin O'Connell's +2.3 per season (95% CI: [+1.3, +3.2]) is entirely positive despite only three~~

Tab. 1: Top 15 Coaches by Average WAR per Season (minimum 3 seasons). Confidence intervals are computed from the t-distribution applied to each coach's individual-season WAR values (Equation 5). Intervals are marginal 95% CIs intended for descriptive ranking; they are not adjusted for the implicit family of pairwise comparisons among coaches and should not be used for hypothesis testing between named coaches. Values reflect Pipeline 2 estimates (full-data refit).

Rank	Coach	Seasons	Total WAR	Total 95% CI	Avg. WAR	Avg. 95% CI
1	John Madden	9	+20.7	[+12.7, +28.7]	+2.3	[+1.4, +3.2]
2	Kevin O'Connell	3	+5.8	[−0.2, +11.7]	+1.9	[−0.1, +3.9]
3	Mike McDaniel	3	+5.6	[−1.9, +13.1]	+1.9	[−0.6, +4.4]
4	John Ralston	5	+9.0	[+5.5, +12.4]	+1.8	[+1.1, +2.5]
5	Kevin Stefanski	5	+8.8	[−2.5, +20.0]	+1.8	[−0.5, +4.0]
6	Nick Sirianni	4	+6.3	[−0.6, +13.2]	+1.6	[−0.2, +3.3]
7	Don Shula	26	+39.6	[+30.1, +49.0]	+1.5	[+1.2, +1.9]
8	Kyle Shanahan	8	+12.2	[+3.6, +20.7]	+1.5	[+0.5, +2.6]
9	Bill Parcells	19	+25.7	[+14.3, +37.1]	+1.4	[+0.8, +2.0]
10	Gene Stallings	4	+5.4	[−5.5, +16.3]	+1.4	[−1.4, +4.1]
11	Mike Vrabel	6	+8.1	[+0.1, +16.1]	+1.3	[+0.0, +2.7]
12	Mike Martz	6	+8.0	[−1.1, +17.0]	+1.3	[−0.2, +2.8]
13	Chuck Pagano	6	+7.7	[−8.3, +23.6]	+1.3	[−1.4, +3.9]
14	Red Miller	4	+4.7	[−5.6, +15.1]	+1.2	[−1.4, +3.8]
15	Matt Nagy	4	+4.6	[+0.4, +8.7]	+1.1	[+0.1, +2.2]

Tab. 2: Bottom 15 Coaches by Average WAR per Season (minimum 3 seasons). Confidence intervals are computed as in Table 1. Intervals are marginal 95% CIs intended for descriptive ranking; they are not adjusted for the implicit family of pairwise comparisons among coaches and should not be used for hypothesis testing between named coaches. Values reflect Pipeline 2 estimates (full-data refit).

Rank	Coach	Seasons	Total WAR	Total 95% CI	Avg. WAR	Avg. 95% CI
1	Steve Spagnuolo	3	−9.6	[−21.1, +2.0]	−3.2	[−7.0, +0.7]
2	Pat Shurmur	4	−12.4	[−20.1, −4.7]	−3.1	[−5.0, −1.2]
3	Scott Linehan	3	−8.9	[−21.8, +4.0]	−3.0	[−7.3, +1.3]
4	Matt Eberflus	3	−8.7	[−17.5, +0.1]	−2.9	[−5.8, +0.0]
5	Rod Marinelli	3	−7.6	[−18.5, +3.3]	−2.5	[−6.2, +1.1]
6	Jack Patera	7	−17.8	[−31.8, −3.7]	−2.5	[−4.5, −0.5]
7	Paul Wiggin	3	−7.3	[−16.2, +1.7]	−2.4	[−5.4, +0.6]
8	David Shula	5	−11.5	[−27.6, +4.5]	−2.3	[−5.5, +0.9]
9	Bruce Coslet	8	−18.1	[−27.1, −9.1]	−2.3	[−3.4, −1.1]
10	J.D. Roberts	3	−6.4	[−12.7, −0.1]	−2.1	[−4.2, −0.0]
11	Abe Gibron	3	−6.2	[−19.4, +7.0]	−2.1	[−6.5, +2.3]
12	Ed Biles	3	−6.1	[−17.0, +4.8]	−2.0	[−5.7, +1.6]
13	Hue Jackson	4	−7.7	[−18.5, +3.1]	−1.9	[−4.6, +0.8]
14	Mike Nolan	4	−7.3	[−16.9, +2.2]	−1.8	[−4.2, +0.5]
15	Jim Schwartz	5	−8.8	[−17.8, +0.3]	−1.8	[−3.6, +0.1]

~~seasons, while coaches such as Gene Stallings and Red Miller have intervals that span zero, suggesting their point estimates may be influenced by individual standout seasons. Total WAR intervals capture both per-season impact and career length: Shula's +52.1 wins (95% CI: [+40.1, +64.0]) represent the most precisely estimated career in the dataset.~~John Madden's +2.3 WAR per season (95% CI: [+1.4, +3.2]) over 9 seasons, Don Shula's +1.5 per season (95% CI: [+1.2, +1.9]) over 26 seasons, and Bill Parcells's +1.4 per season (95% CI: [+0.8, +2.0]) over 19 seasons all have intervals entirely above zero. Shorter careers naturally produce wider intervals: Kevin O'Connell's +1.9 per season (95% CI: [−0.1, +3.9]) and Mike McDaniel's +1.9 (95% CI: [−0.6, +4.4]) reflect just three seasons each, with intervals that narrowly span zero. Total WAR intervals capture both per-season impact and career length: Shula's +39.6 wins (95% CI: [+30.1, +49.0]) represent the most precisely estimated long-career mark in the dataset.

Approximately ~~3.7~~3.8 wins per season separates coaches at the 95th percentile from those at the 5th percentile, consistent with the three-to-four victory range reported by Hadley et al. (2000).

4.2 Coach WAR in Aggregate

Figure 1 illustrates the relationship between coaching longevity and WAR performance by plotting coaches based on average WAR per season and career length (measured in seasons coached).

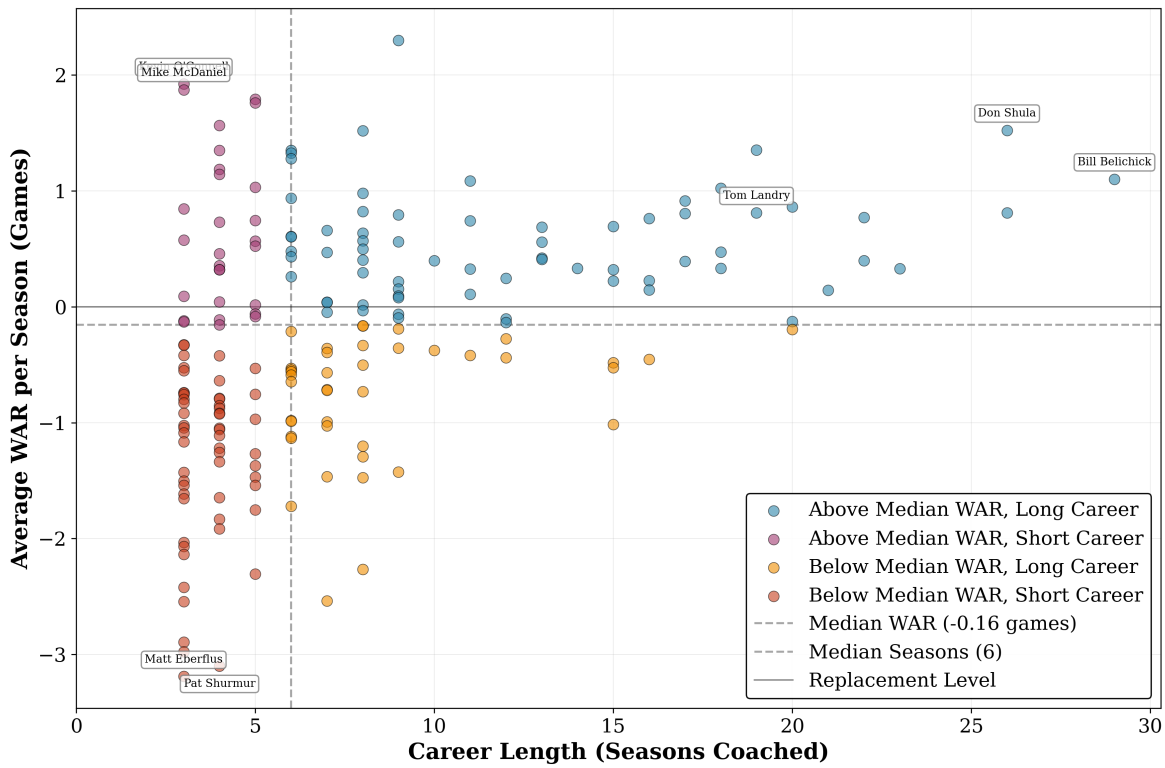

Fig. 1: Coach WAR Career Distributions. Coaches with two or fewer seasons are excluded.

The upper-right and lower-left quadrants show a positive relationship between coaching quality and career length, consistent with organizations retaining successful coaches. However, the lower-right quadrant contains coaches with 5+ seasons despite near-zero or negative WAR, suggesting possible suboptimal retention decisions. Context-adjusted evaluation may help identify such cases earlier.

4.3 Offensive versus Defensive Coaches

A recurring question concerns whether offensive or defensive coaching backgrounds produce more successful head coaches. For this analysis, coaches were classified based on their coordinator or position coaching experience prior to their head coaching appointment. ~~Across the full modern era (1970–2024), no significant aggregate difference emerges between the two groups.~~ Across the full modern era (1970–2024), the aggregate difference between the two groups is small in magnitude and the corresponding 95% confidence interval includes zero. However, this aggregate view obscures substantial era-level variation. Table 3 reports mean WAR by background and decade alongside Mann–Whitney U tests with 95% confidence intervals on the difference.

Tab. 3: Mann-Whitney U Test comparing Coaching Backgrounds by Decade. 95% CIs are Welch's t-intervals on the difference in means (Offensive − Defensive).

Metric	1970's	1980's	1990's	2000's	2010's	2020's
Coaches (O/D)	34/21	30/22	33/28	40/37	42/33	33/20
Seasons (O/D)	87/82	146/83	143/126	152/151	158/135	92/56
Mean WAR (O/D)	−0.472/+0.434	+0.174/+0.106	−0.335/+0.195	−0.380/−0.105	−0.149/−0.015	+0.396/−0.068
Difference (O − D)	−0.906	+0.068	−0.530	−0.275	−0.134	+0.464
95% CI	[−1.42, −0.39]	[−0.41, +0.55]	[−0.92, −0.14]	[−0.65, +0.10]	[−0.58, +0.32]	[−0.17, +1.09]
U-statistic	2416	6088	7484	10496	10188	2889
p-value	0.0003	0.953	0.017	0.199	0.510	0.217

In the 1970s, defensive-background coaches ~~held a significant advantage: the offensive-minus-defensive difference was −0.87 wins per season (95% CI: [−1.45, −0.29], $p = 0.005$)~~outperformed offensive-background coaches by 0.91 wins per season (offensive-minus-defensive difference −0.91; 95% CI: [−1.42, −0.39]; $p = 0.0003$). By the 2000s, the groups were ~~indistinguishable~~statistically indistinguishable (~~−0.08~~−0.28, 95% CI: [~~−0.49~~−0.65, +~~0.34~~0.10], $p =$ ~~0.86~~0.20). In the 2020s, the direction has reversed, with offensive coaches averaging +~~0.59~~0.46 WAR per season more than defensive coaches (95% CI: [~~−0.05~~−0.17, +~~1.24~~1.09], $p =$ ~~0.11~~0.22). ~~The 2020s interval includes zero, indicating that this reversal, while suggestive, has not yet reached conventional significance levels with the available sample. However, if the current performance differential persists through the remainder of the decade, the roughly doubled sample size would yield a projected $p \approx 0.02$, reaching conventional significance.~~ The 2020s 95% CI for the difference includes zero, so the reversal should be read as a directional pattern rather than as a confirmed effect at the present sample size.

Cumulatively, offensive coaches have improved by +~~0.83~~0.87 wins per season from the 1970s to the 2020s (95% CI: [+~~0.28~~0.36, +~~1.38~~1.38], $p =$ ~~0.004~~0.0009), while defensive coaches have declined by ~~−0.63~~−0.50 over the same period (95% CI: [~~−1.30~~−1.14, +~~0.04~~0.13], $p =$ ~~0.064~~0.12).

4.4 ~~WAR Persistence and Scheme Obsolescence~~Year-to-Year Persistence of Coaching WAR

We find evidence that Coach WAR lacks year-to-year persistence.~~Coaching performance is unstable from year to year.~~ Analyzing 1,163 consecutive season pairs from 1970–2023, Year N coaching WAR explains only ~~6.9~~6.4% of the variance in Year N+1 performance (Figure 2). Top-quintile coaches averaging +~~2.58~~2.32 WAR in Year N regress to +~~0.54~~0.61 in Year N+1, retaining roughly ~~21~~26% of their advantage. Bottom-quintile coaches at −~~2.79~~2.60 WAR improve to ~~−1.47 WAR~~−1.33 WAR after applying a 10th percentile penalty for mid-season terminations (Figure 3).

This volatility contrasts sharply with raw winning percentage, which shows approximately double the year-to-year stability ($R^2 = 0.137$, Appendix D, Figure 4). Winning percentage likely captures persistent organizational advantages, such as market size, ownership stability, and accumulated roster talent, that carry forward regardless of coaching. The coach-specific contribution, by contrast, fluctuates considerably from season to season.

~~One interpretation of this low persistence is scheme obsolescence: coaching innovations that produce short-term advantages may be studied, copied, or countered by opposing staffs, reducing the advantage over time. Shared game film and a competitive market for coordinators and assistants provide plausible mechanisms for such diffusion. However, because WAR already conditions on roster composition and injury context, the residual volatility is specifically in the coaching component, though the inherent noise of a 16–17 game season limits the precision of single-year estimates.~~

Most of this single-season volatility reflects irreducible sampling noise rather than genuine season-to-season swings in coaching quality. As noted in Section 3.5.1, a 16-game season carries a binomial sampling

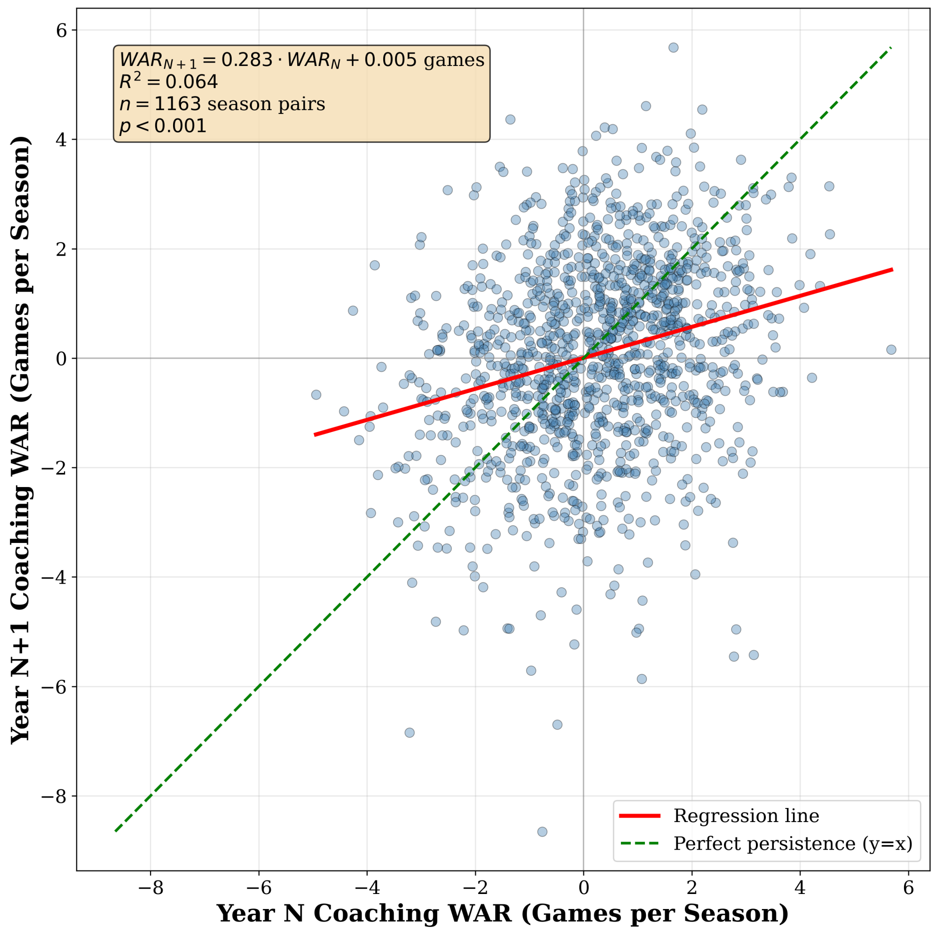

Fig. 2: Coach WAR in Year N+1 vs. Year N (persistence)

floor of approximately 1.94 wins even for a model with perfect knowledge of each team's underlying win probability, and the model's test RMSE of roughly 2.5 wins sits only modestly above it. A single season is therefore an inherently noisy measurement of coaching quality, much as a single season of batting average on balls in play is a noisy measure of pitching skill (McCracken, 2001). Low year-to-year autocorrelation is the expected consequence and does not by itself imply that a persistent coaching signal is absent: the signal is simply small relative to season-level noise. Because that noise is independent across seasons, averaging over a coach's career cancels much of it while preserving the underlying signal, which is why the career-level estimates that anchor our conclusions are far more reliable than any single-season value.

A more speculative possibility is that part of the genuine, non-noise component of coaching value also erodes over time through scheme obsolescence: innovations that yield short-term advantages are studied, copied, or countered by opposing staffs. A familiar player-level analogue is the quarterback "sophomore slump," in which a standout rookie often regresses in his second season once defensive coordinators have a full year of film to study his tendencies, the player-level counterpart of scheme obsolescence. We stress that this is a multi-year mechanism, not a claim that strategic change undoes a coach's skill within a single season, and that the autocorrelation alone cannot separate scheme obsolescence from sampling noise. What pure noise would not produce, however, is a systematic difference between offensive- and defensive-background coaches, which we examine next.

The instability is not uniform across coaching backgrounds. While both offensive and defensive coaches show similar single-year persistence ($R^2 =$ ~~0.064~~0.047 and ~~0.092~~0.101 respectively), multi-year averaging reveals a divergence. At a 2-year horizon, defensive coaches' predictive power approximately doubles to

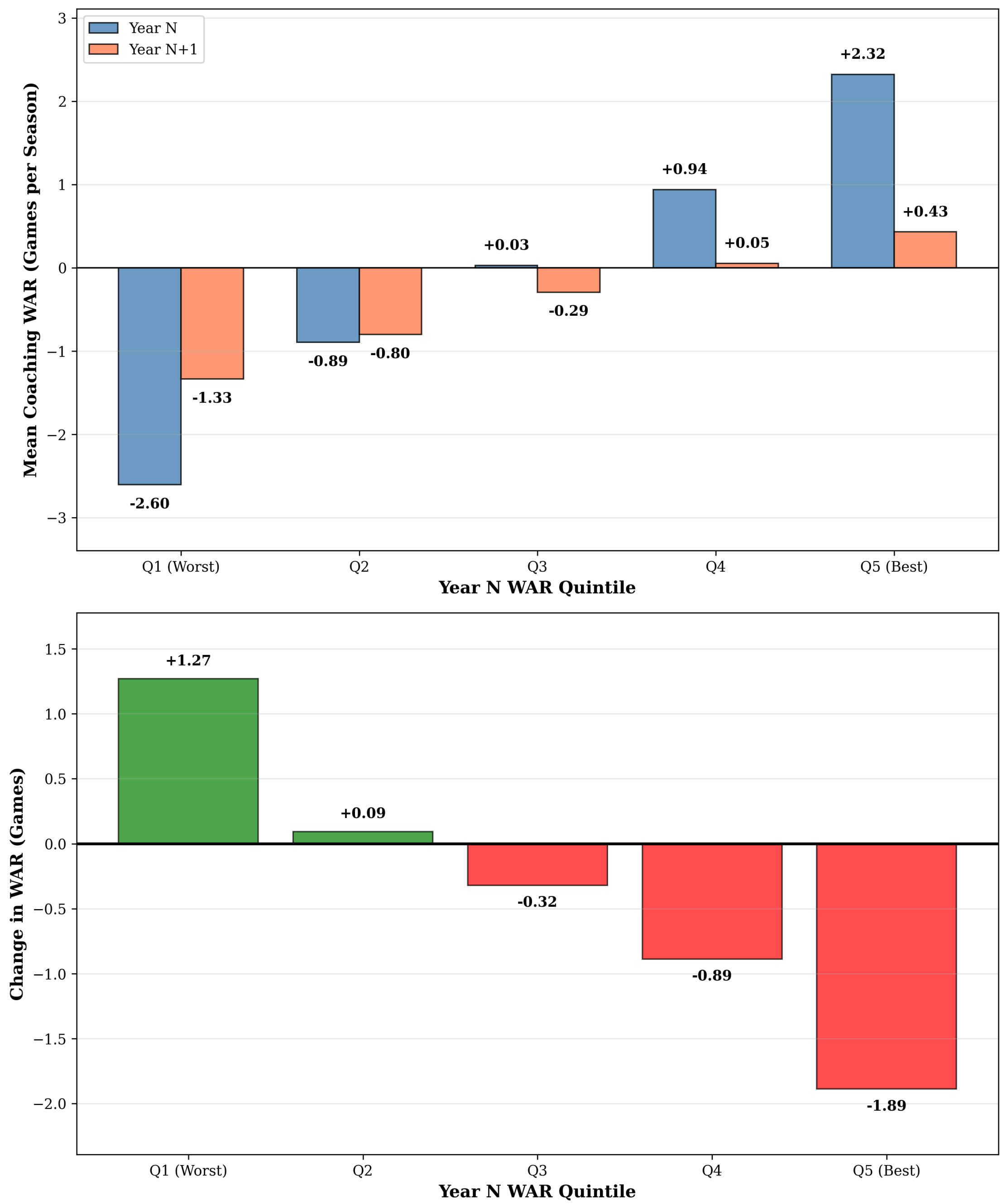

Fig. 3: Regression to the mean by WAR quintile. Top panel: mean WAR in Year N (blue) and Year N+1 (orange) for each quintile. Bottom panel: mean change in WAR from Year N to Year N+1. Bottom-quintile Year N+1 values apply a 10th percentile penalty for coaches terminated mid-season.

$R^2 = \cancel{0.180}0.183$, while offensive coaches remain flat at $R^2 = \cancel{0.069}0.051$. ~~This difference is statistically significant (Fisher's Z $p = 0.009$;~~ The Fisher's Z test on the difference between the two correlations yields $p = 0.001$ (Appendix E, Figure 5 and Table 9). At three-year horizons, the gap persists ($p = \cancel{0.020}0.042$). ~~These patterns suggest that offensive schemes may depreciate more rapidly than defensive approaches, consistent with the notion that offensive innovations are more easily studied and countered from game film.~~This asymmetry is strongest at the two-year horizon, the one persistence test that survives the Benjamini–Hochberg correction (Section 3.5.5). Because the multi-year horizons are built from overlapping rolling averages and are therefore not independent, we read the pattern as tentative, suggestive evidence, rather than a demonstration, that offensive schemes may depreciate more rapidly than defensive approaches.

Of the 13 inferential tests described in Section 3.5.5, five survive the Benjamini–Hochberg adjustment at $q < 0.05$ ($p^* \approx 0.019$): the overall persistence Pearson test, the 1970s and 1990s decade Mann–Whitney comparisons, the 1970s-vs-2020s offensive Welch's t-test, and the 2-year-horizon persistence Fisher's Z test. The 3-year-horizon persistence test crosses to marginal status under BH adjustment ($q \approx 0.091$); other tests have larger p-values and are described above using their effect sizes and intervals rather than as confirmed effects.

4.5 Limitations and the "Franchise Quarterback Effect"

Tables 4 and 5 show the top and bottom 10 single-season WAR instances in the modern era.

Tab. 4: Top 10 Single-Season WAR Instances. Model residual standard error: ±2.5 wins per season. Values reflect Pipeline 2 estimates (full-data refit).

Rank	Team	Year	Coach	Actual Win %	Replacement Win %	WAR	±1 SE
1	CHI	1987	Mike Ditka	0.733	0.378	+5.7	[+3.2, +8.2]
2	DET	2024	Dan Campbell	0.882	0.594	+4.6	[+2.1, +7.1]
3	RAI	1976	John Madden	0.929	0.644	+4.5	[+2.0, +7.0]
4	CLT	2009	Jim Caldwell	0.875	0.591	+4.5	[+2.0, +7.0]
5	MIN	2017	Mike Zimmer	0.812	0.537	+4.4	[+1.9, +6.9]
6	PIT	2004	Bill Cowher	0.938	0.665	+4.4	[+1.9, +6.9]
7	CLE	2023	Kevin Stefanski	0.647	0.383	+4.2	[+1.7, +6.7]
8	SFO	2019	Kyle Shanahan	0.812	0.551	+4.2	[+1.7, +6.7]
9	KAN	2024	Andy Reid	0.882	0.626	+4.1	[+1.6, +6.6]
10	RAV	2019	John Harbaugh	0.875	0.619	+4.1	[+1.6, +6.6]

Tab. 5: Bottom 10 Single-Season WAR Instances. Model residual standard error: ±2.5 wins per season. Asterisk (*) denotes mid-season termination. Values reflect Pipeline 2 estimates (full-data refit).

Rank	Team	Year	Coach	Actual Win %	Replacement Win %	WAR	±1 SE
1	ATL	2020	Dan Quinn*	0.000	0.541	−8.7	[−11.2, −6.2]
2	SEA	1982	Jack Patera*	0.000	0.428	−6.8	[−9.3, −4.3]
3	PHI	1971	Jerry Williams*	0.000	0.423	−6.8	[−9.3, −4.3]
4	CIN	1996	David Shula*	0.143	0.562	−6.7	[−9.2, −4.2]
5	DAL	2010	Wade Phillips*	0.125	0.491	−5.9	[−8.4, −3.4]
6	CLT	1972	Don McCafferty*	0.200	0.557	−5.7	[−8.2, −3.2]
7	HTX	2020	Bill O'Brien*	0.000	0.341	−5.5	[−8.0, −3.0]
8	CLT	2011	Jim Caldwell	0.125	0.464	−5.4	[−7.9, −2.9]
9	DET	1976	Rick Forzano*	0.250	0.577	−5.2	[−7.7, −2.7]
10	CLE	1984	Sam Rutigliano*	0.125	0.448	−5.2	[−7.7, −2.7]

The highest single-season WAR values reveal an important limitation. Five of the top 10 single-season values coincide with elite or exceptional quarterback play ~~(Tom Brady with Belichick in 2017, Peyton Manning with Caldwell in 2009, Brett Favre with Holmgren in 1998, rookie Ben Roethlisberger's 15–1 season with Cowher in 2004, and rookie Andrew Luck's turnaround with Pagano in 2012)~~(Ken Stabler with Madden in 1976, Peyton Manning's MVP season with Caldwell in 2009, rookie Ben Roethlisberger's 15–1 season with Cowher in 2004, Patrick Mahomes with Reid in 2024, and Lamar Jackson's MVP season with

Harbaugh in 2019). This "franchise quarterback effect" reflects the inherent challenge of isolating coaching contributions: elite quarterbacks and elite coaching are rarely independent. While the framework controls for roster quality through player age, experience, and positional spending, it cannot fully disentangle whether a coach maximizes quarterback talent or whether an elite quarterback elevates apparent coaching performance. The remaining top-10 entries—~~Ditka with Jim McMahon starting only 6 of 15 games (1987), Campbell with Jared Goff (2024), Shanahan with Jimmy Garoppolo (2019), Stallings with Gary Hogeboom (1989), and Zimmer with backup Case Keenum (2017)~~Ditka with an injury-limited Jim McMahon (1987), Campbell with Jared Goff (2024), Shanahan with Jimmy Garoppolo (2019), Stefanski with a journeyman group led by Joe Flacco (2023), and Zimmer with backup Case Keenum (2017)—occurred without consensus elite quarterbacks, indicating the framework captures coaching performance beyond quarterback quality. Career-average analysis with confidence intervals (Tables 1–2) naturally mitigates single-season noise.

Nine of the ten bottom single-season WAR values (Table 5) correspond to mid-season terminations, with win percentages based only on games coached before dismissal. Scaling partial-season records to a full 16-game season may amplify negative WAR estimates, though the firing itself reflects an organizational judgment that performance was unsustainable.

Bill Walsh's career illustrates both genuine coaching development and single-season variability: his first season (1979, 2–14) produced ~~−4.8~~−4.3 WAR, yet his best season (1987) reached +~~3.5~~3.2 WAR—a ~~8.4~~7.5-win swing that preceded a third Super Bowl victory. Walsh's early struggles giving way to a dynasty reflect that coaching ability is not static. The trajectory also illustrates the limits of separating coach quality from concurrent player development: Joe Montana joined the 49ers as a third-round rookie in 1979, became the full-time starter in 1981, and reached his statistical peak in the mid-to-late 1980s, while Jerry Rice arrived in 1985. The model partially accounts for this through position-level features such as tenure, age, and salary cap share, which evolve as starters develop and absorb part of the resulting contribution. The attribution remains imperfect because the model conditions on roster-level features rather than identifying individual players, so the WAR estimated for Walsh during this period reflects some of the joint development of the coach and his franchise quarterback. This blending is not entirely a flaw: head coaches contribute to player development through staff hiring, scheme design, and personnel decisions, so attributing a portion of a quarterback's growth to the coach is defensible. The same consideration applies to other coach–quarterback pairings discussed in this section. ~~The model residual standard error of ±2.1 wins per season (Section 3.5) indicates that individual-season WAR values carry substantial uncertainty. Career averages with confidence intervals, as presented in Tables 1–2, provide more reliable assessments and should be preferred for evaluation purposes.~~

Additional limitations warrant consideration. First, while the model controls for roster quality through age, experience, turnover, and salary allocation, these proxy measures cannot fully capture player talent; teams with similar demographic profiles may differ substantially in talent level. Second, organizational factors beyond our data, such as ownership stability, drafting competence, and facility quality, likely influence outcomes but remain unobserved.

4.6 Practical Applications

Context-adjusted coaching evaluation may inform hiring, retention, and contract decisions by separating coaching performance from inherited roster quality. For example, a coach with a .500 record but consistently positive WAR may be undervalued relative to a coach with a .600 record who inherited superior talent. Similarly, retention decisions benefit from objective performance baselines: cumulative negative WAR over multiple seasons, particularly when the confidence interval excludes zero (Table 2), provides quantitative evidence that performance deficits are structural rather than circumstantial.

5 Conclusion

This paper introduces a counterfactual framework for evaluating NFL head coach performance by comparing actual team outcomes to predictions under replacement-level coaching. Three principal findings emerge. First, coaching impact is substantial: approximately ~~3.7~~3.8 wins per season separate coaches at the 95th percentile from those at the 5th percentile, with career leaders ranging from +2.3 WAR per season (95% CI: [+~~1.3~~1.4, +3.2]) to ~~−3.3~~−3.1 (95% CI: [~~−4.7~~−5.0, ~~−2.0~~−1.2]). Second, the relative effectiveness of offensive and defensive coaching backgrounds has shifted markedly over the modern era, from a ~~significant~~ defensive advantage in the 1970s (~~−0.87~~−0.91, 95% CI: [~~−1.45~~−1.42, ~~−0.29~~−0.39]) to a suggestive offensive advantage in the 2020s (+~~0.59~~0.46, 95% CI: [~~−0.05~~−0.17, +~~1.24~~1.09]). Third, coaching WAR exhibits low year-to-year persistence ($R^2 =$ ~~0.069~~0.064), with top-quintile coaches losing ~~approximately 79%~~much of their Year N advantage by Year N+1, ~~consistent with scheme obsolescence as opponents adapt~~driven primarily by unobserved single-season variance with secondary contributions from gradual scheme adaptation. Offensive coaches show ~~significantly less multi-year persistence than defensive coaches (Fisher's Z $p = 0.009$ at the 2-year horizon).~~ less multi-year persistence than defensive coaches: at the 2-year horizon, offensive $R^2 = 0.051$ versus defensive $R^2 = 0.183$ (Fisher's Z test, $p = 0.001$).

Among several directions for future work, two appear particularly promising. First, replacing draft pick counts with expected value estimates based on draft position would better capture roster talent, as a first-overall selection carries substantially more value than a late-round pick within the same round. Second, a game-level WAR methodology, analogous to the play-level approach of Yurko et al. (2019), would enable bootstrap resampling for model-based confidence intervals on single-season point estimates rather than relying on residual-based approximations.

Data Availability

Coaching history data was scraped from Pro Football Reference. Processed data and analysis code are available at https://github.com/jwilliamson7/Coach_WAR or upon request to the corresponding author. Raw salary cap data from Spotrac is proprietary but processing scripts are provided.

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A Feature Count Across Categories

Tab. 6: Feature Count Across Categories

Feature Category	Coach Feature?	Number of Features
Metadata (excluded from training)	No	2
Salary Cap	No	19
Roster Turnover	No	63
Starter Turnover	No	27
Injury/Availability	No	37
Player Demographics	No	44
Penalty/Opponent	No	7
Draft Capital	No	29
Coaching Experience	Yes	7
Prior Team Performance (OC)	Yes	33
Prior Team Performance (DC)	Yes	33
Prior Team Performance (HC)	Yes	66
Target Variable	No	1
Total Columns in Dataset		**368**
Features Used in Training		**365**

B Selected Hyperparameters for the Final Model

Tab. 7: Selected Hyperparameters for the Final Model

Hyperparameter	**Value**
Objective	reg:squarederror
Random State	42
Number of Estimators	300
Learning Rate	0.05
Max Estimator Depth	2
Gamma	0
Lambda	0.1
Alpha	0.1
Subsample	0.7
Colsample by Tree	1.0
Minimum Child Weight	5

C Feature Importance Ranking of Top 20 Features

Tab. 8: Feature Importance Ranking of Top 20 Features

Rank	Feature Description	Importance	Coach Metric
1	Active 53-man roster cap as percentage	0.045	No
2	Reserve/injured/suspended cap as percentage	0.041	No
3	Normalized opponent total yards under HC	0.037	Yes
4	Normalized opponent rushing TDs under HC	0.027	Yes
5	Normalized opponent interceptions thrown	0.021	No
6	Average percentage of games missed by QBs	0.019	No
7	Normalized team interceptions thrown	0.019	No
8	Normalized points scored under HC	0.018	Yes
9	Average experience of all starters	0.017	No
10	Normalized opponent rushing yards under HC	0.015	Yes
11	Normalized opponent points scored under HC	0.014	Yes
12	Normalized red-zone attempts under HC	0.012	Yes
13	Normalized opponent net yards per pass attempt under HC	0.010	Yes
14	Number of prior head-coaching hires	0.010	Yes
15	Normalized opponent rushing first downs under HC	0.010	Yes
16	Average experience of all roster players	0.009	No
17	Normalized scoring percentage under HC	0.009	Yes
18	Normalized avg points per drive under HC	0.008	Yes
19	QB new player rate percentage	0.008	No
20	Kicker cap as percentage	0.008	No

D Win Percentage Persistence

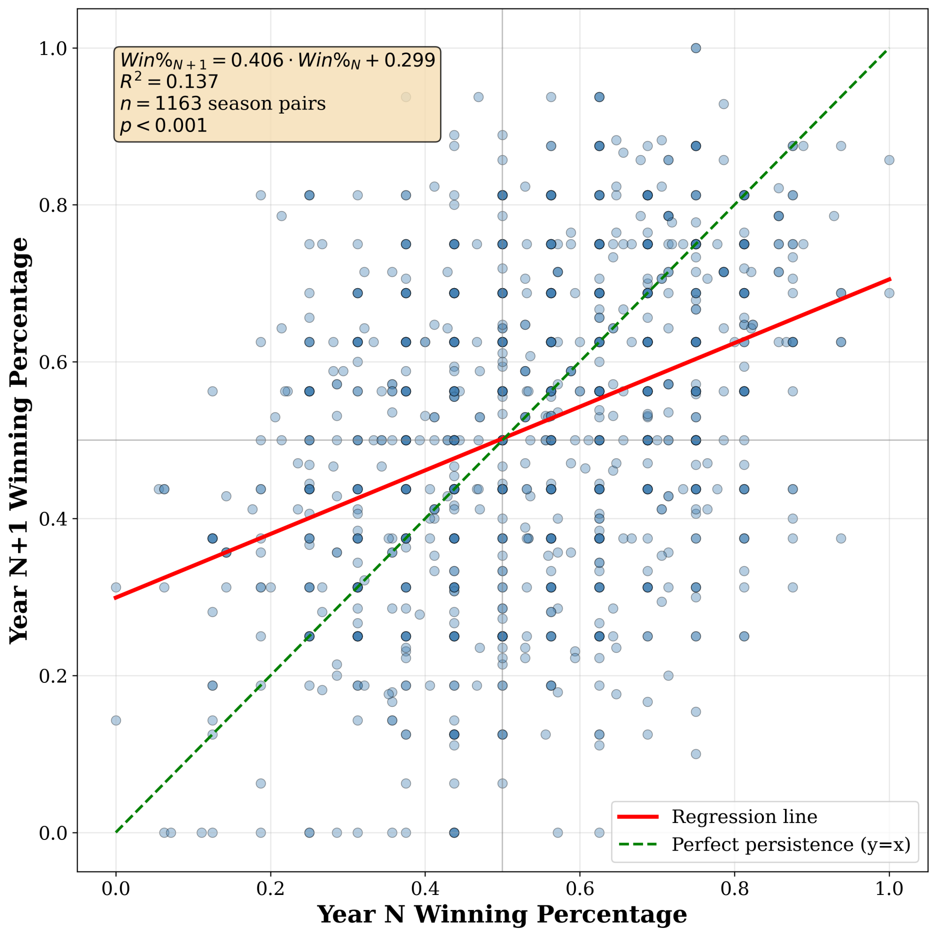

Fig. 4: Coach Win Percentage in Year N+1 vs. Year N

E WAR Persistence by Coach Background

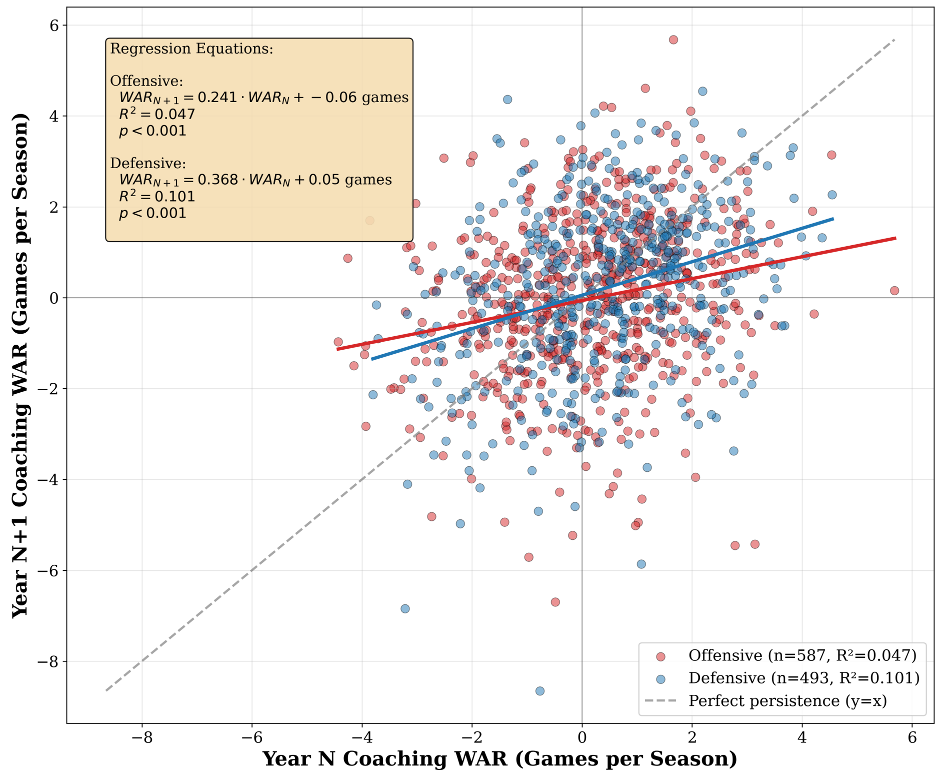

Fig. 5: Coach WAR in Year N+1 vs. Year N by Coach Background

Tab. 9: Coaching WAR Persistence by Background and Time Horizon

	Offensive		Defensive		Other	
Time Horizon	**R^2**	**p-value**	**R^2**	**p-value**	**R^2**	**p-value**
1-year	0.047	<0.001	0.101	<0.001	0.007	0.454
2-year avg	0.051	<0.001	0.183	<0.001	0.001	0.790
3-year avg	0.053	<0.001	0.146	<0.001	0.006	0.600
4-year avg	0.053	<0.001	0.149	<0.001	0.000	0.987

Note: Fisher's Z-test for difference in correlations (Offensive vs Defensive): 1-year (p=0.074), 2-year (p=0.001), 3-year (p=0.042), 4-year (p=0.073).